

Section A

Answer **all** the questions in this section.

- 1 Enzyme **X** digests a sugar. Fig. 1.1 shows the rate of reaction of enzyme **X** and a mixture of enzyme **X** and substance **Y** on the sugar. Both reactions were carried out in a conical flask.

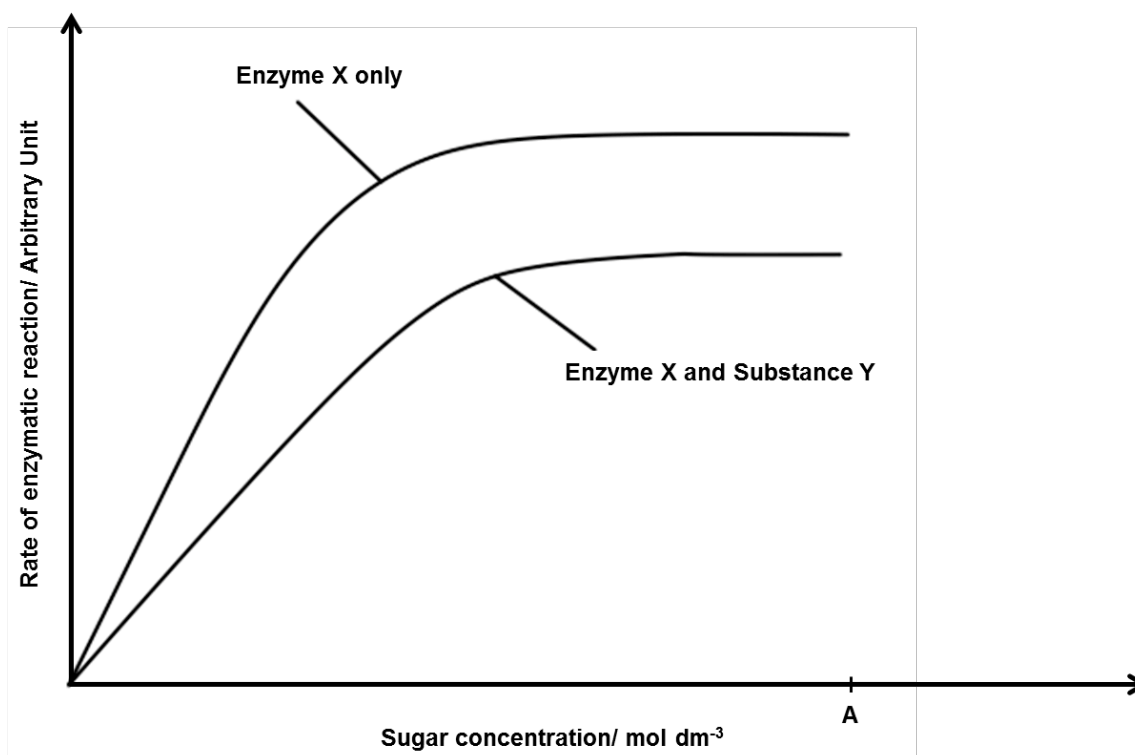


Fig. 1.1

- (a) With reference to Fig. 1.1, describe and explain the effect of substance **Y** on the enzymatic reaction. [3]

- Even at very high substrate concentration, the maximum rate of reaction in the presence of substance **Y** is lower than that in absence of substance **Y**/ Increasing substrate concentration cannot overcome inhibition by substance **Y**/ Maximum rate of reaction $V_{\max}$ is lower and affinity for the sugar is not affected by substance **Y**/ K_m remains unchanged;
- Substance **Y** is a non-competitive inhibitor which has no close structural resemblance to the substrate;
- binds irreversibly to a region other than active site of enzyme **X**;

[1mark each]

- (b) Substance **Y** was found in one of the products of the reaction catalysed by enzyme **X** after the substrate concentration, **A**, in Fig.1.1.

Predict how the trend of the **graph for enzyme X only** would continue after sugar concentrations greater than A. Sketch the part of the graph in Fig. 1.1. [1]

- Decreasing trend due to end-product inhibition (refer to **Fig. 1**).

Enzyme **X** may be immobilised in alginate beads. Fig. 1.2 shows a comparison between the activity of free enzyme **X** and immobilised enzyme **X** over a range of temperatures. Equal concentrations of free enzyme **X** and immobilised enzyme **X** were used.

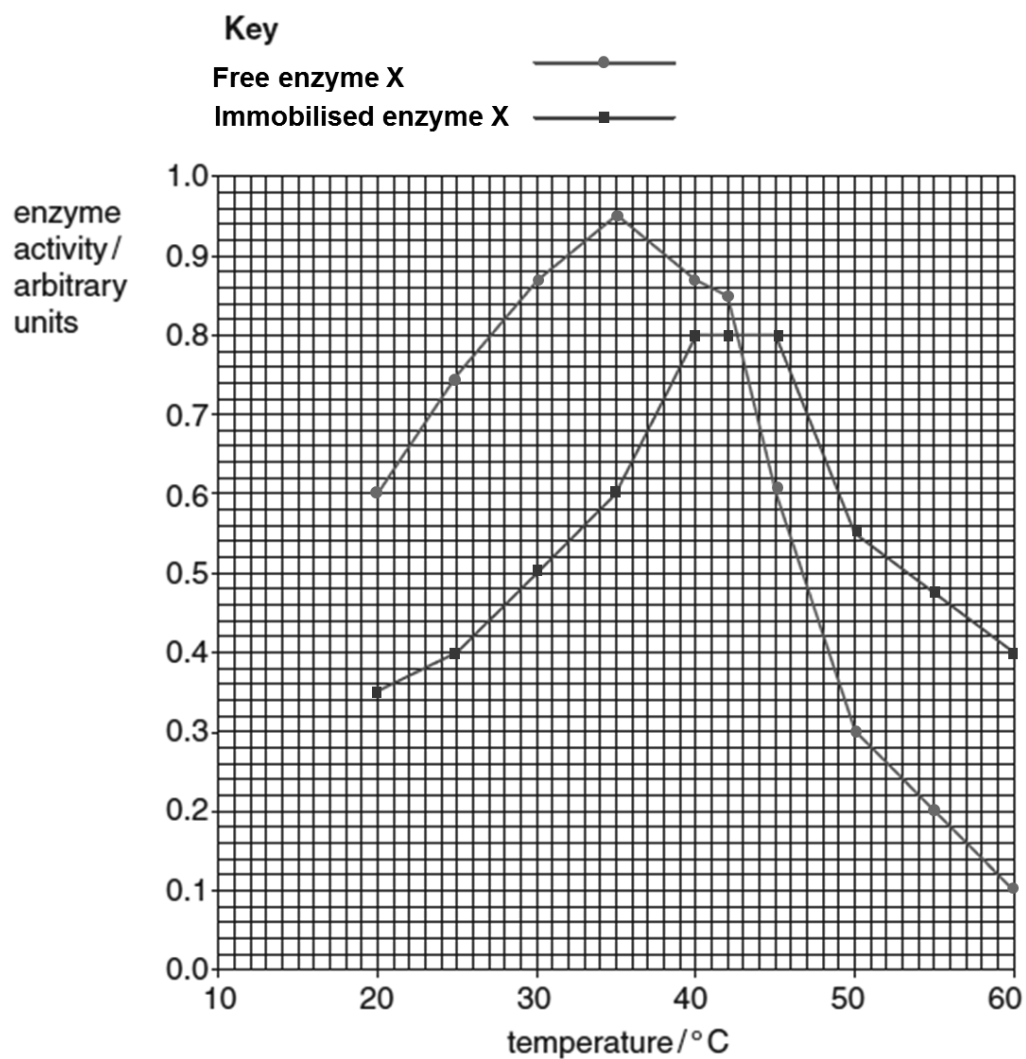


Fig. 1.2

(c) With reference to the Fig. 1.2,

(i) describe and explain the effect of temperature on the activity of free enzyme **X**. [3]

- As temperature increases from 20 to 35 °C, the rate of reaction of free enzyme **X** increases from 0.60 to 0.95 A.U. due to the increased kinetic energy of the sugar substrate and enzyme molecules;
- These molecules move faster colliding with one another in correct orientation to form more enzyme-substrate complexes;
- The increased rate of reaction increases until it reaches its maximum rate of 0.95 A.U. at the optimum temperature of 35 °C when most enzyme-substrate complexes formed;
- Beyond the optimum temperature of 35 °C to 60 °C, rate of reaction decreases from 0.95 to 0.1 A.U. as intramolecular bonds (reference to hydrogen bonds, hydrophobic interactions) are broken and three-dimensional conformation of active sites in enzymes is altered such that substrate can no longer fit

[1 mark each, max 3 marks]

(ii) describe and explain the differences between the activity of free enzyme **X** and immobilised enzyme **X** up to 40 °C. [2]

Description

- From 20 to 40 °C, activity of immobilised enzyme is lower than free enzymes (reference to any suitable data range) and the maximum activity of immobilised enzyme (0.80 A.U. at 35 °C) is lower than that of free enzyme (0.95 A.U. at 40°C/ between 40 to 45 °C);

[1 mark each, max 1 mark]

Explanation

- Enzymes immobilised in alginate beads diffuse out from beads less easily to come into contact with sugar substrate/ less able to move and collide successfully with substrates leading to fewer enzyme-substrate complexes formed;
- Products formed within the beads diffuse out less easily and accumulate leading to end-product inhibition.

[1 mark each, max 1 mark]

The oil stored in the seeds of the castor oil plant *Ricinus communis* has been used in the production of pharmaceutical products such as zinc and castor oil cream, and as a lubricant. However, *Ricinus* seeds also produce a poisonous protein ricin, which is classified as an agent of bioterrorism.

Fig. 1.3 shows the structure of ricin which consists of two polypeptides – the A chain of ricin (RTA, ricin toxin A) joined to a B chain (RTB, ricin toxin B).

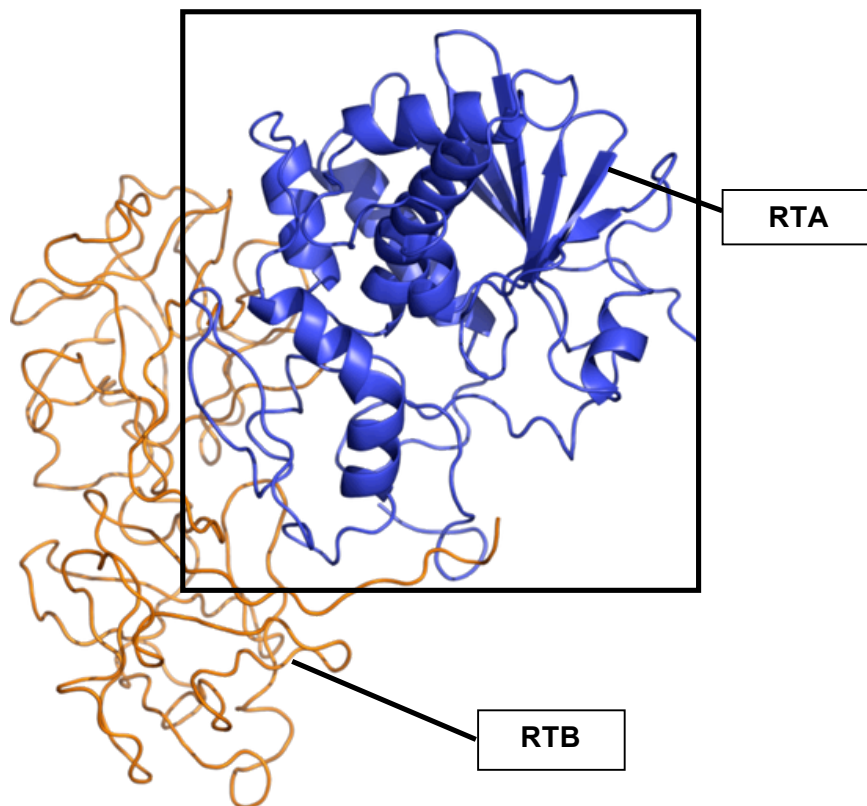


Fig. 1.3

(d) With reference to Fig. 1.3,

- (i) identify the level of organisation of the structure of ricin.
- Quaternary structure

[1]

- (ii) describe how the conformation of RTA (boxed section) is obtained from a linear polypeptide chain. [3]

- The linear polypeptide chain undergo folding to form the secondary structure;
- which is made up of the α -helix and the β -pleated sheet;
- The secondary structure of RTA is maintained by hydrogen bonds formed between the $-\text{CO}$ and $-\text{NH}$ groups of the main amino acid chain;
- *Note: R groups are not involved in the formation of hydrogen bonds in the secondary structure of a polypeptide*;
- The secondary structure undergo further folding and bendings to form the tertiary structure / globular structure of RTA,
- The tertiary structure of RTA is maintained by 4 types of bonds (disulfide bond, hydrogen bond, ionic bond and hydrophobic interaction) that are formed between R groups of RTA.

- (e) The RTB component of ricin has two binding sites specific for galactose. This enables RTB to bind to cell surface components that contain galactose. This is the first step of cellular intoxication. Identify a structure found on the mammalian cell surface that makes it a good target for ricin. [1]
- Glycoprotein

- (f) The RTA component of ricin is an enzyme that damages an essential component of the protein synthesis machinery – the ribosome. RTA catalyses the removal of one particular nitrogenous base (adenine) in a crucially important section of ribosomal RNA of the large subunit of eukaryotic ribosomes. This results in eventual cell death.

The two polypeptides – RTA and RTB are covalently linked by a disulfide bond between two cysteine residues. Reduction of ricin into its subunits, RTA and RTB, is catalysed by protein disulfide isomerase (PDI), an enzyme that resides in the lumen of the endoplasmic reticulum. PDI breaks the interchain disulfide bond.

Fig. 1.4 illustrates the reduction process catalysed by PDI.

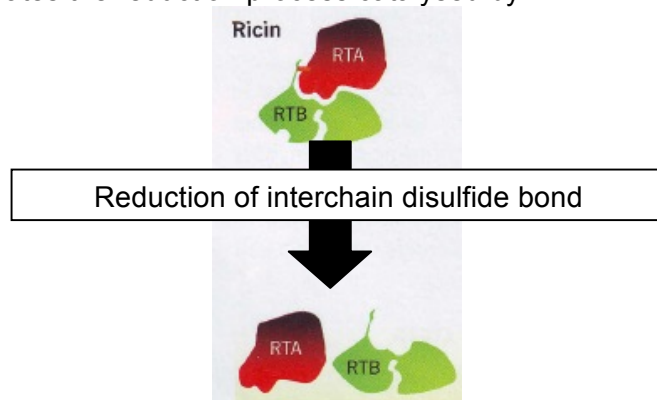


Fig. 1.4

- (i) Explain why the reduction process shown in Fig. 1.4 is necessary before RTA can exhibit its catalytic activity. [2]
- In intact ricin, the RTB polypeptide hides the catalytic site of RTA;
 - Only after RTA is separated from RTB (after the reduction process) does RTA exhibit its catalytic activity.
- (ii) Suggest one way in which plant ribosomes in the seed cells are protected from the effect of ricin. [1]

Either one:

- Ricin is made as an inactive protein;
- It is stored in the plant vacuoles, away from its substrates;

[Total: 17]

- 2 (a) Fig. 2 is an electron micrograph of an organelle found in eukaryotic cells.

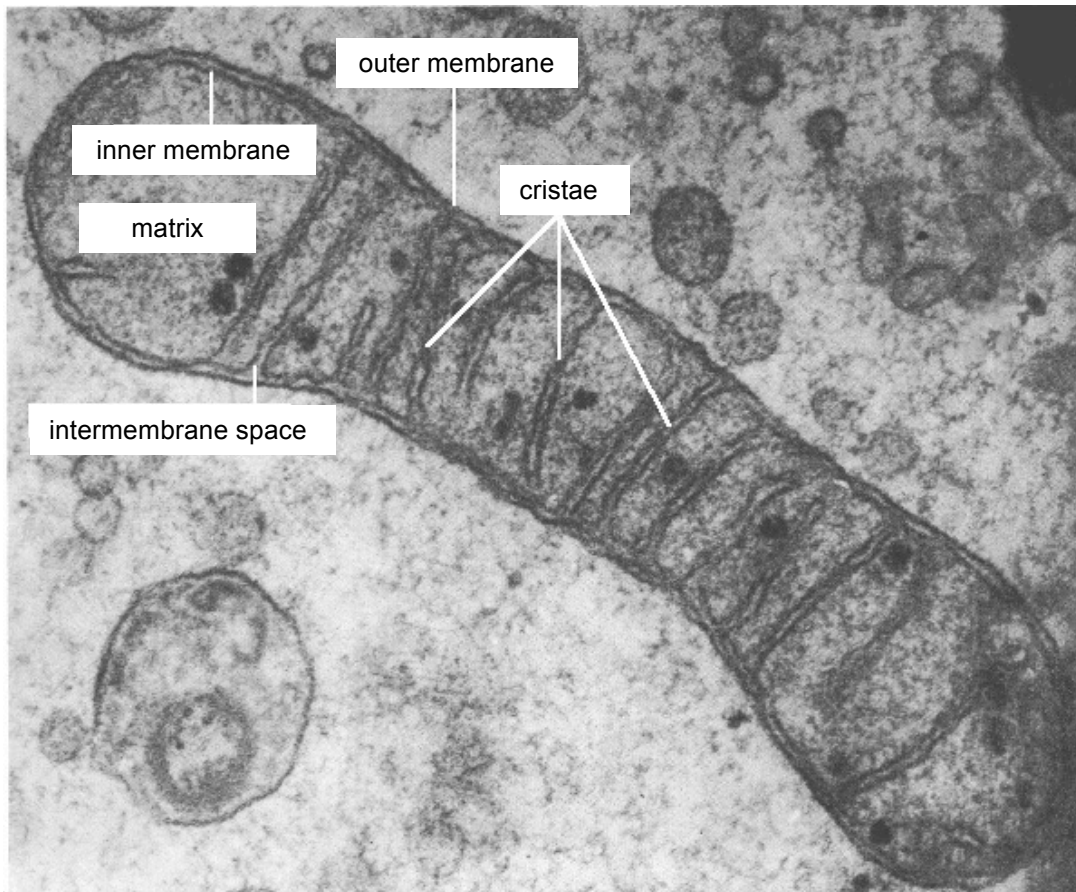


Fig. 2

State **two** similarities between the organelle above and a typical prokaryotic cell like *E. coli*.

[2]

- Both do not have a true nucleus
- Both contain circular DNA
- Both have 70S ribosomes

[Max 2 marks]

- (b) (i) Mutants that are *lac Y*⁻, meaning that the *lac Y* gene is inactivated, retain the capacity to synthesise β -galactosidase. However, even though the *lac I* gene encoding for the repressor protein is still intact, β -galactosidase can no longer be induced by adding lactose to the medium. Explain why this is so. [4]

- The *lac Y* gene codes the enzyme permease which enables cells to take up lactose efficiently. [1]
- Mutants that are *lac Y*⁻ do not have the enzyme permease → cells cannot take up lactose which will thus not be converted to allolactose; [1]
- Which is required to bind to the allosteric site of the lac repressor protein to inactivate it; [1]

[max 2m]

- Hence the repressor protein continues to occupy the operator site on the lac operon; [1]
- RNA polymerase is blocked from accessing the promoter site, thus the structural gene which encodes β -galactosidase cannot be transcribed. [1]

- (ii) Positive control of the *lac* operon results from breakdown of glucose, which alters the levels of cyclic adenosine monophosphate (cAMP) in the cell. cAMP binds to a catabolite activator protein (CAP), forming a CAP-cAMP complex which increases the affinity of RNA polymerase for the *lac* promoter. However, this mechanism will not cause upregulation (genetic regulation that leads to an increase in gene expression) of the *lac* operon in the absence of lactose. Explain why this is so. [2]

- This is because in the absence of lactose, the repressor protein will be bound to the operator; [1]
- and will prevent the binding of RNA polymerase to the promoter. Transcription cannot proceed even though glucose is being actively metabolised. [1]

- (iii) Suggest a reason how a mutation in the operon can result in a repressor that continually binds to the operator. [1]

- There is a mutation in the *lac Z*, *lac Y* or *lac A* gene leading to a dysfunctional enzyme (mention the enzyme). Thus allolactose is not formed and the repressor protein remains bound to the operator. (Any one of the gene is accepted).

Reject: mutation to *Lac I* gene (lies outside the operon)

(c) Why is any bacterial gene capable of being transferred during generalised transduction? [3]

- During generalised transduction, the phage infects the cell and the bacterial chromosome is fragmented. [1]
- While viral particles are being assembled, any one of the bacterial chromosome gene fragments can possibly be incorporated within the viral protein coat. [1]
- This is because it is the amount of DNA and not the information content of the DNA that governs viral particle formation. [1]

Max 2 marks

- When the new phage carrying the bacterial gene infects another cell, it brings with it the new bacterial gene. [1]

[Total: 12]

- 3 The human papillomavirus (HPV) is a small non-enveloped virus that has a double-stranded circular DNA genome. Some HPVs have been well established as the main risk factor of cervical cancers.

The HPV genome is replicated in the nucleus of infected cells in the form of a circular DNA (episome). The HPV E2 gene encodes a transcription factor E2 that suppresses transcription of HPV E6 and E7, which are viral oncogenes. In certain instances during the replication cycle of HPV, the viral DNA can be linearised and integrated into chromosomal DNA of host cells. A breakpoint that disrupts the HPV E2 gene will prevent the synthesis of E2 proteins that normally regulate the transcription of E6 and E7.

Fig. 3.1 illustrates the integration of HPV DNA into the host chromosome.

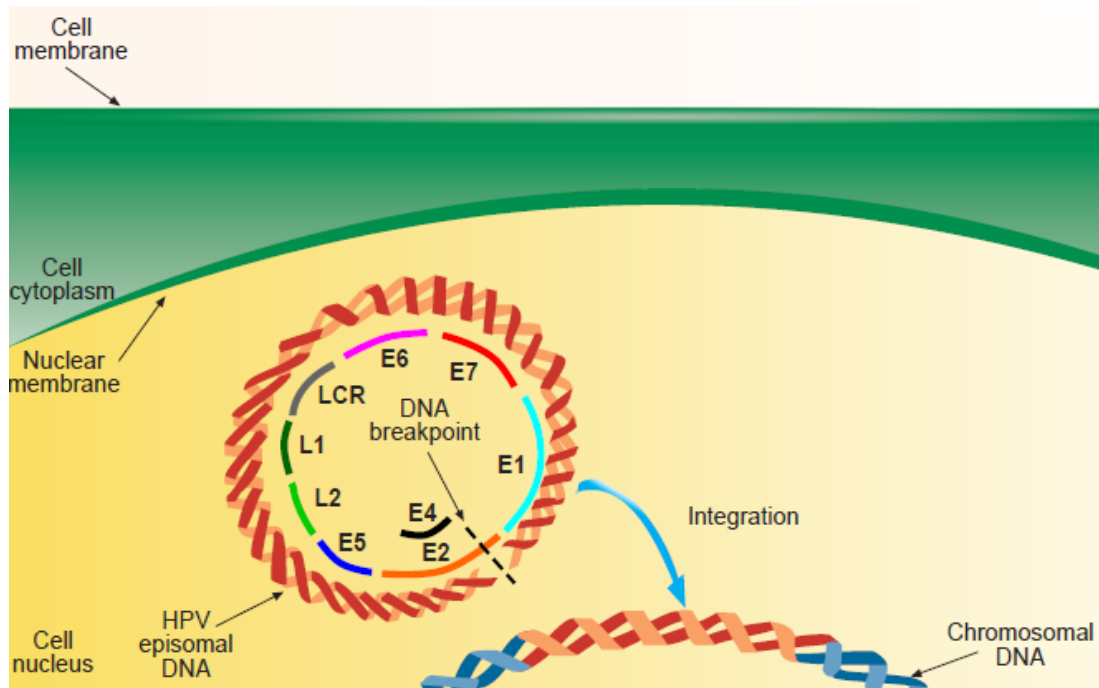


Fig. 3.1

Part of the HPV genome consisting of the long control region (LCR), E6 gene and E7 gene is integrated into the host DNA (Fig 3.2). The LCR contains a variety of regulatory sequences that will regulate viral replication and gene expression. The HPV E6 and E7 genes are consistently expressed, whereas the remaining HPV genes are often deleted or not transcribed after integration.

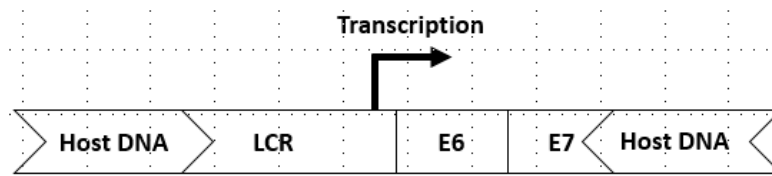


Fig. 3.2

- (a) With reference to the information provided and Fig. 3.2, suggest how a dual-cistronic E6-E7 mRNA transcript is produced. [2]

- The LCR region contains a promoter sequence;
- this allows general transcription factors and RNA polymerase to bind to the promoter to form the transcription-initiation complex to initiate transcription;
- The E6-E7 genes are located adjacent to each other, and will thus be under the control of the same promoter; this results in the formation of a dual-cistronic mRNA following transcription;

[max 2]

An increase in the synthesis of E6 and E7 oncoproteins is associated with the risk of development of carcinoma. Fig. 3.3 illustrates the HPV inactivation of two cellular tumour suppressor proteins, p53 and pRb.

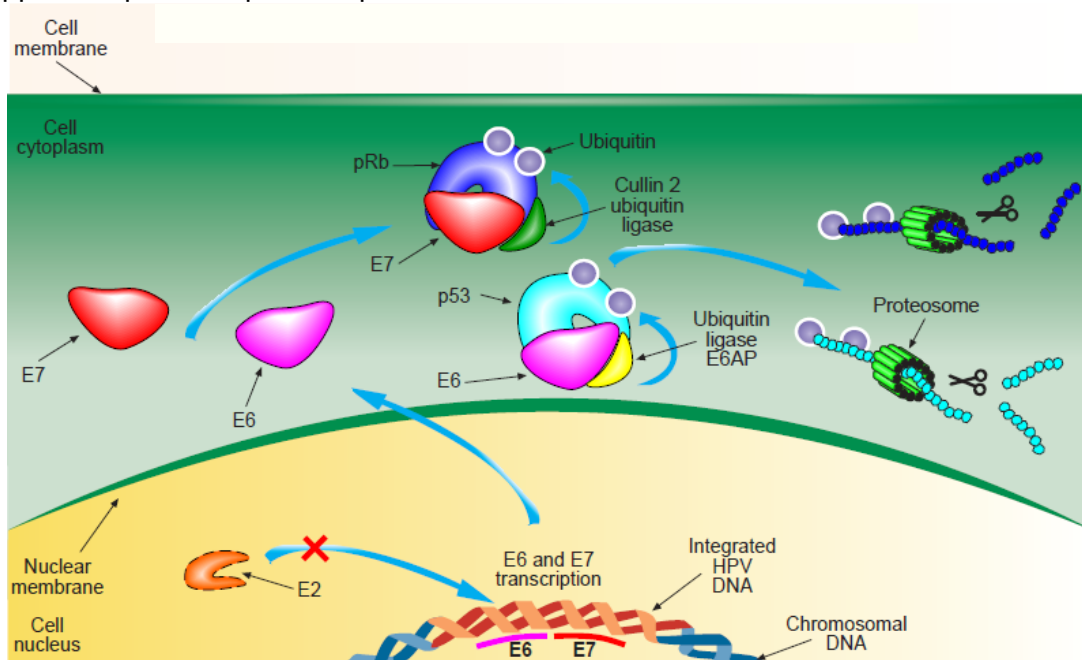


Fig. 3.3

- (b) With reference to Fig. 3.3, describe how the absence of E2 and hence an increase in E6 and E7 protein synthesis, can result in the degradation of p53 and pRb. [2]

- E6 binds to p53 in the cytosol and results in the binding of E6AP ubiquitin ligase, which ubiquitinates p53 which promotes its degradation in the proteasome;
- E7 binds to pRb in the cytosol and recruits cullin 2 ubiquitin ligase, which ubiquitinates pRb which promotes its degradation in the proteasome;

(c) Explain why the degradation of p53 may lead to tumour formation. [3]

- p53 is a transcription factor and its production is induced by DNA damage and abnormalities in chromosomal attachment;
- the role of p53 includes the following (mention either 1):
 - activates genes that promote DNA repair;
 - activates genes that arrest cell division;
 - activates genes that promote apoptosis;
- degradation of p53 will allow a cell that has DNA damage (which in this case is the result of the integration of the HPV genome into host chromosome) to continue to divide and hence increases the risk of tumour formation;

(d) The ends of eukaryotic chromosomes are protected by identical repeating lengths of DNA called telomeres, which prevent chromosomes from unravelling during cell division. Each time a cell divides, the length of the telomeres shortens and the ability of the cell to divide decreases.

Another function of the HPV E6 proteins is their ability to activate the expression of the catalytic subunit of telomerase (hTERT). Explain the relationship between high levels of telomerase expression and cancer. [4]

- Cells with high levels of telomerase, which adds specific DNA sequence repeats / more specific repeating DNA units OR telomeric repeats; to telomeres / DNA strands in telomere regions at the ends of chromosomes;
- thus elongating telomeres OR preventing telomeres from shortening OR preventing shortening of chromosomes to critical length beyond which cell division can no longer occur;
- do not undergo replicative cell senescence;
- as they proliferate, and their division is not checked by the Hayflick limit;
- therefore they continue to divide uncontrollably;
- and do not undergo apoptosis; contributing to development of cancer.

Accept: telomerase expression is a step in multi-step progression of cancer / more mutations accumulated

Reject: higher chance of cancer

[1m each, max 4]

[Total: 11]

- 4 Wing colour in a butterfly species is determined by two alleles, one coding for red pigment and another coding for yellow pigment.

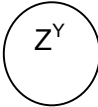
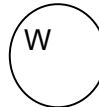
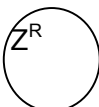
When true breeding yellow butterflies are mated with true breeding red butterflies, the wings of male progeny are always orange. Female progeny always have the same wing colour as their fathers.

- (a) Using suitable symbols, draw **two** different genetic diagrams in the space below to fully explain the results obtained in the progeny. [5]

Z^Y represents Z chromosome with the allele that codes for yellow pigment in wings
 Z^R represents Z chromosome with the allele that codes for red pigment in wings
 W represents W chromosome

[1 mark for correct key]

Scenario 1

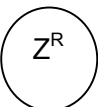
Parental phenotypes	Female, yellow	X	Male, red
Parental genotypes	$Z^Y W$	X	$Z^R Z^R$
Gametes			

1 mark for correct parental phenotype, parental genotype and circled gametes

F_1 genotypes	$Z^Y Z^R$	$Z^R W$
F_1 phenotypes	Male, orange	Female, red (same as father)
F_1 phenotypic ratio	1	: 1

1 mark for correct F_1 genotype, F_1 phenotype and F_1 phenotypic ratio

Scenario 2

Parental phenotypes	Female, red	X	Male, yellow
Parental genotypes	$Z^R W$	X	$Z^Y Z^Y$
Gametes			

1 mark for correct parental phenotype, parental genotype and circled gametes

F_1 genotypes	$Z^Y Z^R$	$Z^Y W$
F_1 phenotypes	Male, orange	Female, yellow (same as father)

F_1 phenotypic ratio	1	: 1
------------------------	---	-----

1 mark for correct F_1 genotype, F_1 phenotype and F_1 phenotypic ratio

(b) Explain the mode of inheritance of wing colour in the butterfly species. [3]

- In the butterfly species, the female is the heterogametic sex. Females with the Z^Y allele ($Z^Y W$) would have yellow wings whereas those with the Z^R allele ($Z^R W$) would have red wings. (Hint: Female progeny always have the same wing colour as their fathers.)
- The male is the homogametic sex. Males with genotype $Z^Y Z^Y$ would have yellow wings and those with genotype $Z^R Z^R$ would have red wings.
- The alleles Z^Y and Z^R exhibit incomplete dominance relative to each other. Hence, males with the genotype $Z^Y Z^R$ have an intermediate phenotype of orange wings.

[1 mark each]

(c) Observed results of the above genetic cross differ from the expected results.

Suggest **two** reasons why such a discrepancy occurs, referring only to events that occur after meiosis. [2]

- Sample size is too small
- Variation is due to chance/ insignificant
- Differential survival of gametes/ non-random mating;
- Differential survival of fertilised zygotes/ some individuals die before being sampled

[1 mark each, max 2 marks]

[Total: 10]

5 Fig. 5 shows the steps involved during glycolysis.

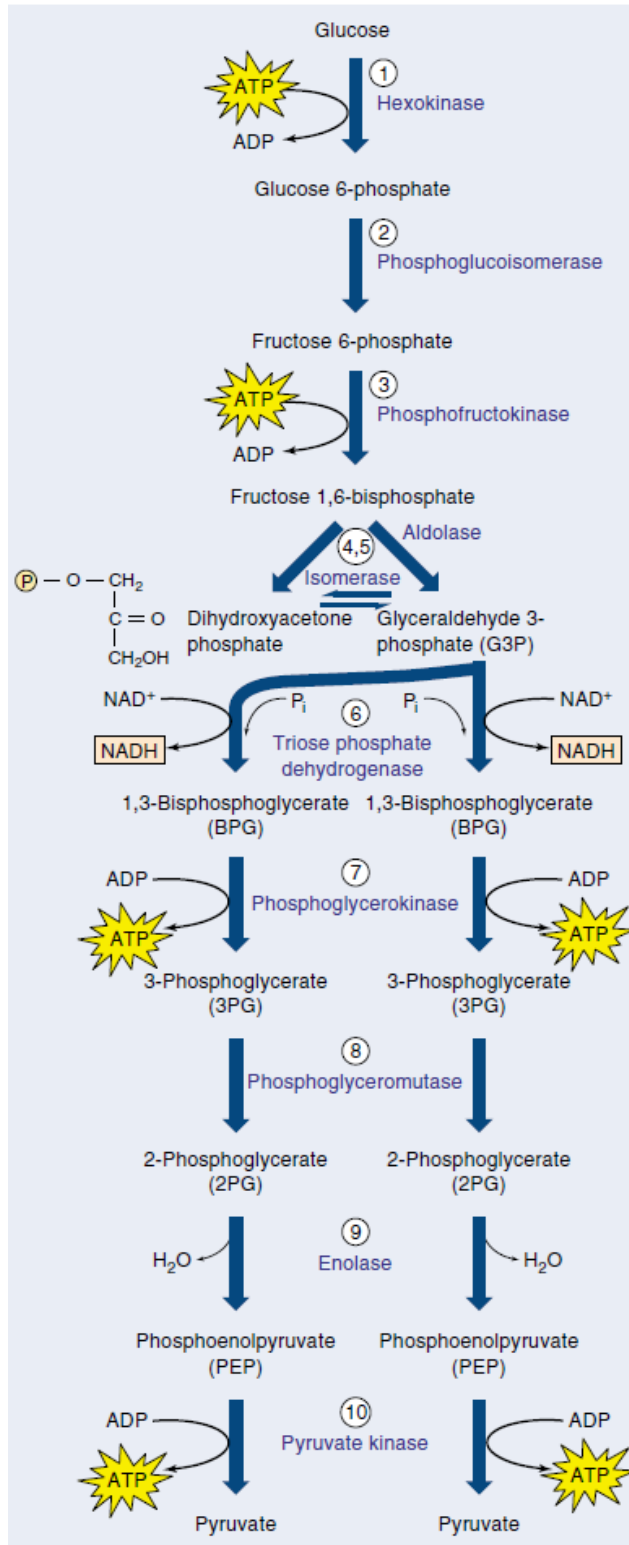


Fig. 5

(a) With reference to Fig. 5,

(i) explain the role of hexokinase in the first step of glycolysis. [2]

- Hexokinase phosphorylates glucose by transferring a phosphate group from ATP to glucose, converting it to glucose-6-phosphate;
- This activates/ raises the energy level of glucose for ATP production in subsequent steps of glycolysis / respiration.

[1 mark each]

(ii) explain the role of glycolysis in aerobic respiration. [2]

- Glycolysis produces a net gain of two ATPs by substrate-level phosphorylation and reduces NAD for oxidative phosphorylation in aerobic respiration (to produce ATP);
- Glycolysis also breaks down one glucose molecule into two molecules of pyruvate that enter into the mitochondrion / link reaction, followed by Krebs cycle where pyruvate is further broken down / completely oxidized to produce more ATP.

[1 mark each]

(iii) describe how a named organism obtains energy for survival in the absence of oxygen from glycolysis. [2]

- Alcoholic fermentation in yeast: Pyruvate broken down to acetaldehyde/ ethanal then to ethanol;
OR
- Lactate fermentation in any named mammal: Pyruvate broken down to lactate;
- NAD^+ regenerated in process to allow glycolysis to continue.

[1 mark each]

- (b) Glyceraldehyde-3-phosphate is an intermediate in glycolysis which is also found in the Calvin cycle in plants.

Describe how glyceraldehyde-3-phosphate is formed in the Calvin cycle in plants. [4]

- RuBP carboxylase (Rubisco) catalyses the fixation of a carbon dioxide molecule with a molecule of ribulose biphosphate (RuBP);
- forming two molecules of glycerate-3-phosphate / 3-phosphoglycerate (GP);
- glycerate-3-phosphate/ 3-phosphoglycerate (GP) utilises ATP from light-dependent reaction to give 1,3-biphosphoglycerate;
- 1,3-biphosphoglycerate is then reduced to glyceraldehyde-3-phosphate (GALP/ PGAL/ G3P/ TP) using NADPH from light-dependent reaction.
(**Reject** use of acronyms without spelling full names.)

[1 mark each]

[Total: 10]

- 6 Cry toxins are produced by the soil bacterium *Bacillus thuringiensis* (Bt). By genetic engineering, Cry toxin gene can be incorporated into commercially important crops to specifically kill insect pests without affecting other organisms.

Cry toxins are believed to cause death in target insects by oligomerising into pore complexes in cell surface membrane of midgut epithelial cells.

- (a) Explain how formation of pores in the cell surface membrane of midgut epithelial cells causes death in target insects. [1]

- The pores allow ions and water to diffuse freely into the cells causing changes in membrane permeability leading to cell swelling/ cell lysis and death of midgut epithelial cells in target insects.

Another mechanism of action of Cry toxin in midgut epithelial cells of target insects has been proposed. Fig. 6 shows the proposed magnesium ion-dependent signalling cascade. Single-pass membrane receptors that cross the membrane only once, such as aminopeptidase N (APN) receptors and cadherin-like receptors, are involved. Cry toxins eventually lead to destabilisation of the cytoskeleton and ion channels in the cell membrane causing cell death.

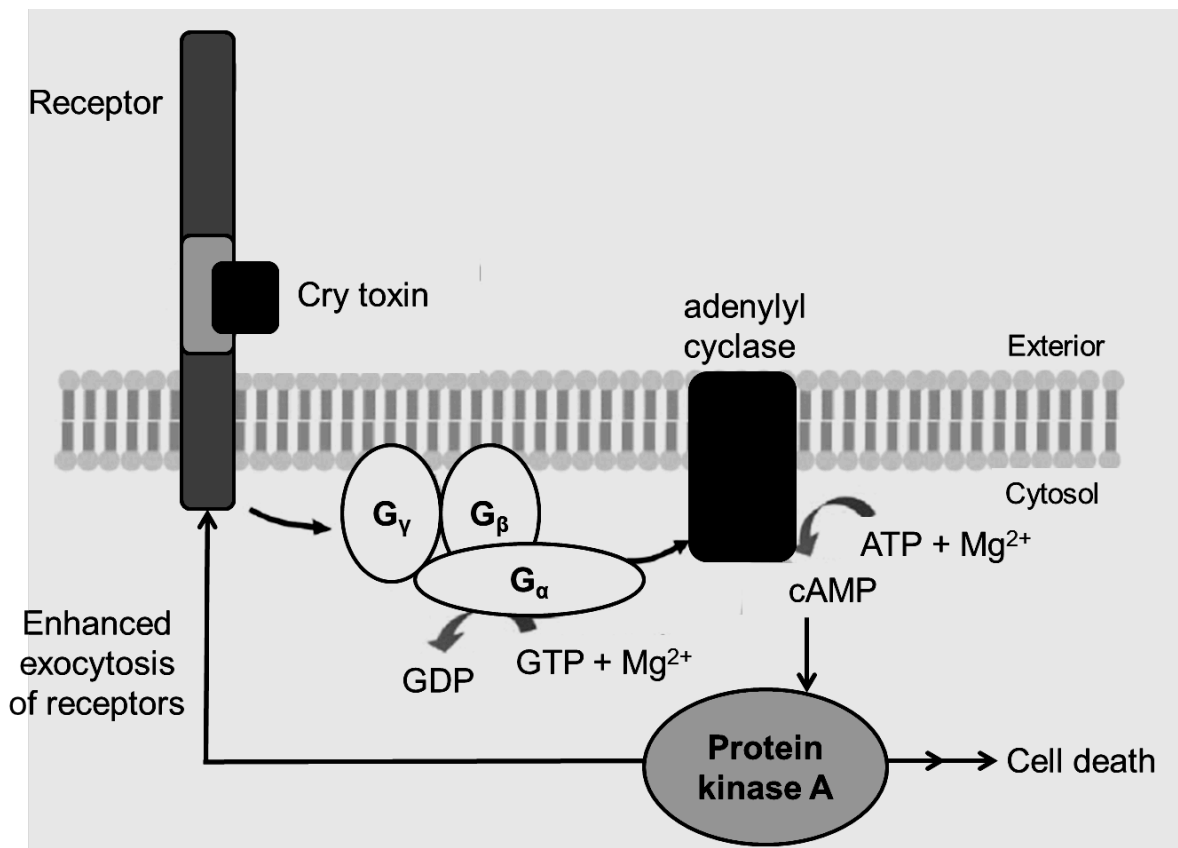


Fig. 6

- (b) (i) With reference to Fig. 6, describe the proposed cell signalling cascade of Cry toxin. [4]

- When Cry toxin (ligand) binds to the receptor, the receptor is activated and cause the membrane-bound G-protein (at the cytoplasmic end) to exchange its GDP with a GTP in the presence of Mg^{2+} ;
(**Reject** Reference to G protein-coupled receptors (GPCR). Receptors of Cry toxins are not GPCR as they only pass the membrane once unlike GPCR that are multi-pass membrane receptors which pass membrane seven times.)
- α -subunit of G protein, G_α , dissociates from its β - and γ - subunits and moves along the membrane to adenylyl cyclase;
- Upon binding of G_α subunit, adenylyl cyclase undergoes conformational change and catalyses conversion of ATP to cyclic adenosine monophosphate (cAMP) in the presence of Mg^{2+} ;
- cAMP acts as a second messenger to activate relay protein, protein kinase A (PKA) which phosphorylates many downstream effector proteins and enzymes in a phosphorylation cascade to induce incorporation of more receptors of Cry toxin at cell surface membrane and to promote cell death.

[1 mark each]

- (ii) State **one** advantage and **one** disadvantage of a signalling cascade. [2]

Advantage

- amplification of signalling cascade: One signal gives rise to multiple cellular responses;

Disadvantage

- Prone to error: error along any part of the cascade would result in undesirable consequences;

OR

Non-specificity of response causes wastage of energy/ resources.

[1mark each]

- (iii) NF449 is an antagonist of G protein that prevents binding of GTP. Suggest how NF449 could affect the signalling pathway of Cry toxins. [2]

- GDP of G protein is not exchanged for GTP inactivating G protein and adenylyl cyclase is not activated leading to a low level of cAMP;
- PKA is inactivated causing reduced cytotoxicity of Cry toxins/ fewer cell death.

[1 mark each]

(c) Suggest why Cry toxins only kill target insects and not other organisms. [1]

- Membrane receptors that bind to Cry toxins are only expressed in the midgut epithelium cells of target insects and not in that of other organisms;
OR
- Conditions of gastrointestinal tract such as pH and ion balance that facilitate activation of Cry signalling cascade in insects are different in other organisms.

[1 mark]

[Total: 10]

- 7 Antifreezes are specialised proteins that prevent the organisms that have them from freezing in the polar oceans of the Arctic and the Antarctic. One of these proteins is the antifreeze glycoprotein (AFGP), which is produced in some fish species. The AFGP proteins protect the fish by lowering their bodies' freezing point temperature so that it is lower than that of the surrounding seawater. This prevents ice crystals from forming within the fishes' body tissues when fish in icy waters drink and feed, which would be lethal.

Fig. 7 shows three species of fish with AFGPs.

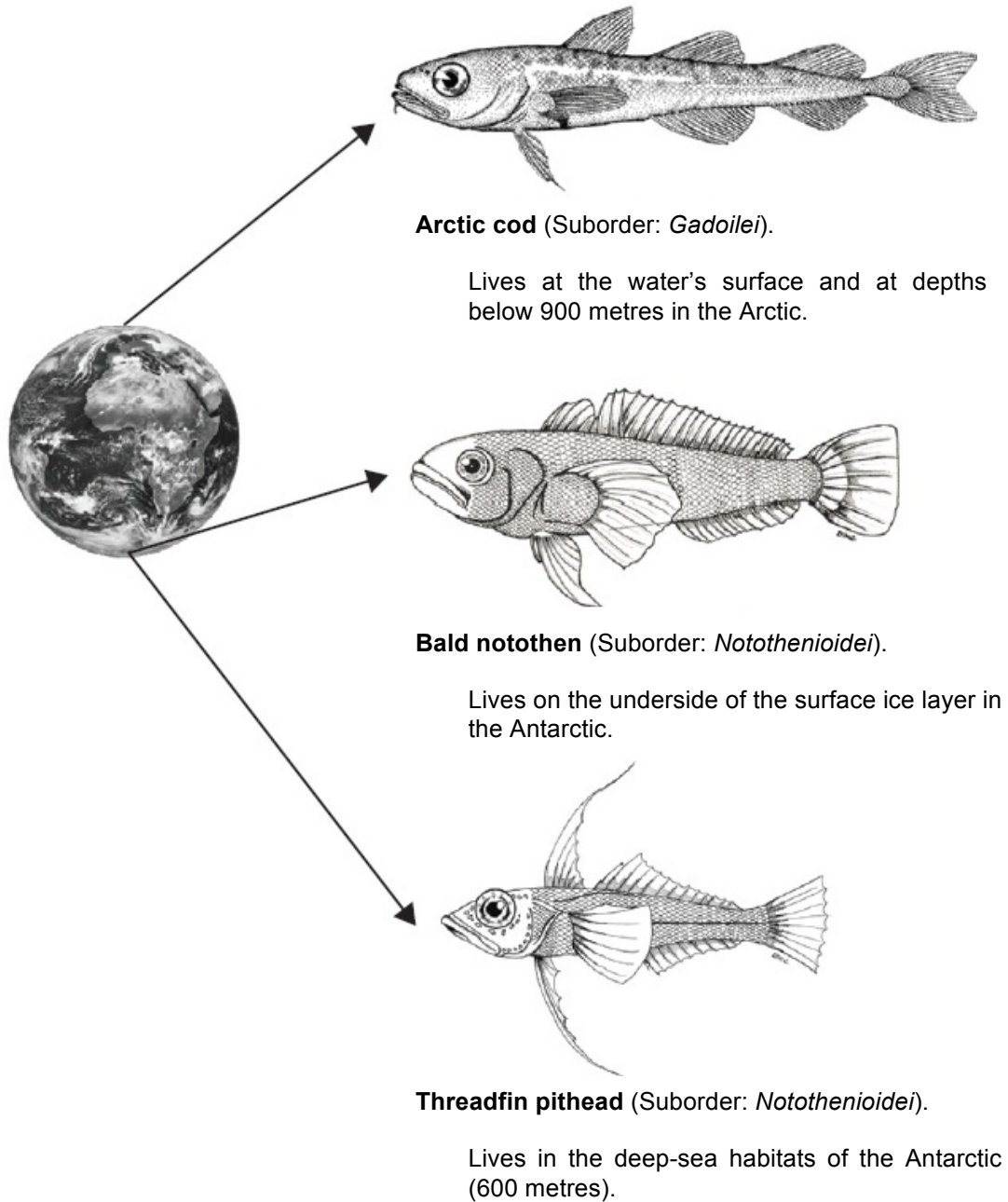


Fig. 7

A circumpolar current contributes to the thermal isolation and low temperatures of present-day Antarctica. This current formed about 30 million years ago when Antarctica separated from South America. Up until this time, the waters around Antarctica had been relatively warm and a wide diversity of fish species flourished.

With the separation of Antarctica from South America, water temperatures began to drop, reaching freezing levels about 10–15 million years ago. At this time, it is thought that the modern day notothenioids were represented by a single species in which the gene for AFGP may have arisen.

Today the seawater temperature around Antarctica seldom varies more than a few degrees above freezing. These waters are now dominated by the notothenioids, of which there are more than 100 species.

AFGP is also found in a species of arctic fish, the Arctic cod. This species is unrelated to the notothenioids, and its AFGP probably evolved about 4–5 million years ago during the glaciation of the Arctic seas. The Arctic cod and the notothenioids separated long before the antifreeze glycoproteins developed.

Researchers have found that the gene for AFGP in the Antarctic notothenioids is quite different in its origin and in its location within the genome from the gene expressing AFGP in the Arctic cod. However, their AFGPs are virtually identical and they both contain the same repeating sequence of three amino acids (threonine-alanine-alanine).

- (a) Explain how natural selection could have brought about the evolution of antifreeze glycoprotein (AFGP) in the notothenioids in the seawater around Antarctica. [5]

- There is variation in the original / ancestral population; some notothenioids having AFGPs and others without;
- A mutation producing the AFGP allele present in the original / ancestral species occurred some time prior to / around the time of the cooling of the waters;
- The colder water / decreasing water temperature acted as a selection pressure on the fish;
- Individuals with the AFGP protein were selected for / were at a selective advantage and are fitter / more likely to survive to sexual maturity and have greater reproductive success;
- to pass on the AFGP allele to their offspring;
- The frequency of the AFGP allele increases in the population;
- Because of mass extinction of fish which lack the AFGP allele / AFGP protein, many new / empty niches became available;
- Speciation of notothenioids occurs, all possessing the AFGP allele;

[1m each, max 5]

(b) (i) What is the term used to describe the evolutionary pattern observed in the Antarctic notothenioids and the Arctic cod? [1]

- Convergent evolution

(ii) Explain how the evolutionary pattern you have identified in **(b)(i)** resulted in the evolution of virtually identical AFGPs in the Antarctic notothenioids and the Arctic cod. [4]

- The antifreeze gene evolved independently as each species arose from different ancestor / unrelated species;
- Analogous features / the same short repeating amino acid sequences found in both Arctic Cod and Antarctic notothenioids results in the expression of the same protein that carry out the same function;
- AFGP genes located in different locations in the genomes of the Arctic Cod and the Antarctic notothenioids indicate that they evolve independently;
- Similar selection pressures on both Arctic Code and in the Antarctic notothenioids resulted in the evolution of the same AFGP molecule;

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where applicable.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a)** Outline the roles of phospholipids, cholesterol and glycolipids in the cell surface membrane. [6]

Phospholipids

1. Phospholipids are arranged in two parallel layers/a phospholipid bilayer ;
2. Hydrophobic core restricts/prevents movement of ions/large polar molecules;
3. Conferring selectivity to membrane;

Cholesterol

4. Cholesterol wedged between phospholipid molecules helps to stabilise the membrane ;
5. Acts like a plug, reducing the escape or entry of small polar molecules across the membrane;
6. Increases flexibility and stability of membranes;
7. At warm temperatures, cholesterol restrains the movement of phospholipids / makes the membrane less fluid;
8. At low temperatures, cholesterol hinders the close packing of phospholipids / makes the membrane more fluid;
9. prevents solidification lowers the temperature required for the membrane to solidify;
10. prevent cell membranes from breaking up;

[Points 1-10, max 4]

Glycolipids

11. Cell-cell recognition;
12. the carbohydrate chain (oligosaccharides) function as a marker that distinguish one cell from another;
13. Cell-cell adhesion;
14. For binding cells together into tissues;
15. Act as receptor sites for chemical signals (e.g. hormones);

[max 2]

(b) Explain how the molecular structure of haemoglobin is related to its function. [7]

<u>Structure</u>	<u>Function</u>
- Each subunit is made up of a protein component called globin and a non-protein component called haem group. - The haem group contains an iron ion (Fe^{2+}) and a porphyrin ring that binds O_2.	- The ability of the haem group to bind to oxygen allows oxygen to be transported in the blood to parts of the body via the circulatory system. Each haem group is able to bind 1 oxygen molecule, there are a total of 4 haem groups in one haemoglobin molecule, so 4 oxygen molecules can be carried at one time → efficient oxygen carrier. - Haemoglobin binds reversibly to oxygen so that it will dissociate to release oxygen to the cells that require oxygen.
The subunits are arranged so that most of its hydrophilic / polar amino acid residues are on the external surface of the globular structure while its hydrophobic amino acid residues are buried in the interior of the globular structure, away from the aqueous surroundings.	Soluble in water and can be transported about in the blood stream. How? Water molecules cluster around their outward-pointing hydrophilic / polar side chains and can form hydrogen bonds with water molecules.
The haemoglobin molecule has a quaternary structure with 4 polypeptide subunits (2 α-chain and 2 β-chain subunits) that are held together by bonds such as hydrogen bonds, hydrophobic interactions and ionic bonds.	The proportion of globin chains are necessary for the normal function of haemoglobin.
Polypeptide subunits up of amino acids that have amino ($-\text{NH}_3^+$) and carboxyl ($-\text{COO}^-$) groups which allow them to act as both an acid and a base.	Amphoteric - Haemoglobin can act as buffer molecules by accepting hydrogen ions, enabling large quantities of carbonic acid to be carried to the lungs without any major alteration in the blood pH (buffering blood pH).

[1 mark per bulleted point, max 7]

(c) Describe the structural differences between collagen and cellulose.

[7]

Collagen	Cellulose
Collagen is a polymer made up of <u>amino acids.</u>	Cellulose is a polymer made up of <u>β-glucose monomers.</u>
Compose of <u>three repeat units</u> of amino acids (Gly-Pro-Hydroxyproline).	Compose of <u>one repeat unit</u> of glucose. (only β-glucose)
Monomers of collagen (amino acids) are linked via <u>peptide bonds.</u>	Monomers of cellulose (β-glucose) are linked via <u>glycosidic bonds.</u>
Collagen is composed of <u>three helical polypeptide chains</u> twisted together to form a <u>triple helix / tropocollagen.</u>	Cellulose chain is made up of <u>a single straight and unbranched chain</u> of β-glucose polymer.
<u>Inter-chain H-bonding</u> maintains the stability of <u>within</u> a single triple helix / tropocollagen. - H-bonding is between <u>-C=O groups of proline residues on one chain and -NH groups of glycine residues on an adjacent polypeptide chain.</u>	- Inter-chain H-bonding maintains the stability <u>between parallel chains of cellulose molecules</u> → this <u>cross-linking</u> by hydrogen bonds form <u>microfibrils.</u> - H-bonding is between <u>OH / hydroxyl groups attached to carbon atom 3 on one chain and carbon atom 6 on a β-glucose on an adjacent chain.</u>
Each tropocollagen <u>crosslinks</u> with a neighbouring tropocollagen running parallel to it → via <u>covalent bonds formed between R groups of lysine residues</u> on adjacent tropocollagen forming <u>collagen fibrils</u>	<u>No covalent cross-links</u> exist between the cellulose microfibrils.

[1 mark per point of comparison]

9 (a) Outline the ways in which mutations can occur. [6]

1. Factors that promotes rate of mutation – exposure to carcinogen, ultraviolet radiation, X-ray, AVP (any 2);
2. Error occurring during replication/repair;

[max 1]

Gene mutation:

3. Gene mutation is defined as the changes in the nucleotide base sequence of the genes;
4. Substitution – replacement of one nucleotide base-pair with another base-pair in a gene;
5. results in missense mutation/nonsense mutation/significant change in the encoded protein/silent mutation/no/little effect on the encoded protein;
6. Insertion – addition of loss of one or more nucleotide base pairs in a gene;
7. Deletion – loss of one or more nucleotide base pair in a gene;
8. results in frameshift mutation, whereby codons are grouped differently;
9. Inversion – a sequence of nucleotides becomes separated from the allele, it rejoins the original position but inverted;

Chromosomal mutation:

10. Chromosomal mutation is defined as the changes in the nucleotide base sequence of the genes or the amount of chromosome;
11. Structural alteration to chromosomes;
12. Translocation – a section of chromosome breaks off and become attached to another chromosome (leading to a new combination of alleles) ;
13. Duplication – (addition of genes) a section of a chromosome replicates (so that a set of gene loci is repeated) ;
14. Inversion – a chromosome breaks at two locations and (the middle portion) flips through 180° before rejoining;
15. Deletion – (loss of genes) a chromosome breaks at two points, the middle portion is displaced with the two ends joining together; (A shorter chromosome containing fewer genes will form).
16. Change in chromosome number;
17. e.g. Aneuploidy – addition or loss of one or more chromosomes (in the nucleus)
or
18. due to non-disjunction during meiosis/ fusion of a normal haploid gamete with a gamete carrying n-2, n-1, n+1 or n+2 chromosomes;
19. Polyploidy - three or more times the haploid number of chromosomes;
20. due to non-disjunction / the fusion of a diploid gamete with a normal haploid gamete giving a triploid nucleus;

[Points 3 – 20: max 5]

(b) Explain how gene mutation leads to cystic fibrosis. [7]

1. Cystic fibrosis (CF) is an inherited disease where the mucus glands are affected.
2. inherited in an autosomal recessive pattern;
3. caused by a mutation of the cystic fibrosis transmembrane conductance regulator (CFTR) gene;
4. found on chromosome 7;
5. a mutation results in deletion of 3 nucleotides CTT;
6. CFTR protein loses an amino acid phenylalanine;
7. at position 508;
8. CFTR protein is a transmembrane chloride channel that allows efflux of chloride ions out of the epithelial cells;
9. mutation of the CFTR gene results in a change in the three-dimensional conformation of the chloride channel;
10. channel is no longer able to transport chloride ions;
11. prevent normal efflux of chloride ions out of epithelial cells;
12. chloride ions accumulate in the epithelial cells, attracting water into the epithelial cells;
13. causing thick and sticky mucus;
14. cells that line the passageways of the lungs, pancreas, and other organs produce thick and sticky mucus.
15. mucus obstruct the airways
16. leading to severe problems with breathing/ resulting in bacterial infections in the lungs/ permanent lung damage/ formation of scar tissue (fibrosis) and cysts in the lungs / causes progressive damage to the respiratory system.
17. mucus blocks the duct of the pancreas/ interferes with the function of the pancreas
18. preventing enzymes and hormones from reaching the intestines to aid digestion and assimilation, leading to digestive problems;
19. reduces life expectancy;

[max 7]

(c) Describe the role of mitosis in maintaining genetic stability. [7]

1. Mitosis maintains genetic stability by ensuring that the two daughter cells contain exactly the same number of chromosomes as the parent cell.
2. Genetically identical
3. The chromatin of the parental nucleus replicate during S phase of interphase before mitosis begins.
4. All chromosomes align at the equator during metaphase...
5. ...to ensure that the sister chromatids are shared equally between the two daughter nuclei after anaphase occurs.
6. The division of the centromere ensures that one sister chromatid of each chromosome will move to opposite poles.
7. There is no association between homologous pairs of chromosomes / bivalents do not form, therefore no chiasmata are formed at prophase I.
8. There is thus no crossing over which would have resulted in genetic variation/recombination of alleles.

[max 7]